Cisco PE Node MAR-PE-13 Optical Transport Failure SOP (Marathahalli Core)

Node: MAR-PE-13 | **Vendor:** Cisco | **Severity:** Critical | **Alarm Code:** ALARM_6650

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve a critical Optical Transport Failure on Cisco PE Node MAR-PE-13 located at the Marathahalli Core site. The failure is characterized by Alarm Code ALARM_6650, indicating a potential SFP/XFP degradation or fiber cut, resulting in Rx/Tx power loss. The PE node's Optical Transport Failure is attributed to the underlying hardware mechanics of the SFP/XFP transceivers, which are responsible for transmitting and receiving light signals over the fiber optic cables. In this failure domain, the SFP/XFP transceivers are prone to degradation due to exposure to harsh environmental conditions, such as high temperatures, humidity, or physical stress, ultimately leading to a decrease in their optical power output. Furthermore, the Optical Transport Failure can also be caused by fiber cuts or breaks, which can occur due to various factors such as physical damage, aging of the fiber, or environmental factors like earthquakes or flooding. The PE node's control plane, which is responsible for managing the routing and switching of data packets, is also affected by this failure, as it relies on the Optical Transport layer for communication with other nodes in the network. In the event of an Optical Transport Failure, the PE node's control plane may become unstable, leading to network instability and potential service outages.

Pre-requisite Checks

- Check: Verify that the remote hands have been dispatched to the Marathahalli Core site to access the PE node., Description: Ensure that the remote hands have the necessary clearance and equipment to access the PE node and perform the required actions.

- Check: Verify the status of the redundant link on the PE node., Description: Ensure that the redundant link is operational and can take over in the event of a failure.

Potential Root Causes

Cause	Probability
SFP/XFP degradation due to exposure to harsh environmental conditions.	High
Fiber cut or break due to physical damage or aging.	Medium
Rx/Tx power loss due to SFP/XFP transceiver failure.	Low

Diagnostic Commands & Expected Outputs

```
Command: show controllers t3/0/0/0, Expected Output: SFP/XFP is present and operational, Description: Verify the presence and operational status of the SFP/XFP transceiver on the T3/0/0/0 interface.
```

```
Command: show interfaces gigabitethernet 0/0/0, Expected Output: Rx/Tx power levels are within normal range, Description: Verify the receive and transmit power levels on the GigabitEthernet 0/0/0 interface.

Command: show log | include ALARM_6650, Expected Output: Alarm code ALARM_6650 is present in the logs, Description: Verify the presence of the Alarm code ALARM_6650 in the system logs.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP transceiver on the T3/0/0/0 interface., Command: no controllers t3/0/0/0, Description: Physically replace the SFP/XFP transceiver on the T3/0/0/0 interface with a new one and configure the interface to use the new transceiver.

Step 2: Step: Verify the redundant link is operational., Command: show interfaces gigabitethernet 0/0/1, Description: Verify that the redundant link on the GigabitEthernet 0/0/1 interface is operational and can take over in the event of a failure.

Step 3: Step: Verify the receive and transmit power levels on the GigabitEthernet 0/0/0 interface., Command: show interfaces gigabitethernet 0/0/0, Description: Verify that the receive and transmit power levels on the GigabitEthernet 0/0/0 interface are within normal range.

Post-Check Verification

```
Command: show log | include ALARM_6650, Expected Output: Alarm code ALARM_6650 is no longer present in the logs, Description: Verify that the Alarm code ALARM_6650 is no longer present in the system logs.

Command: show controllers t3/0/0/0, Expected Output: SFP/XFP is present and operational, Description: Verify the presence and operational status of the SFP/XFP transceiver on the T3/0/0/0 interface.

Command: show interfaces gigabitethernet 0/0/0, Expected Output: Rx/Tx power levels are within normal range, Description: Verify the receive and transmit power levels on the GigabitEthernet 0/0/0 interface.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver on the T3/0/0/0 interface., Command: controllers t3/0/0/0, Description: Restore the original SFP/XFP transceiver on the T3/0/0/0 interface and configure the interface to use the original transceiver.

- Step: Verify the redundant link is operational., Command: show interfaces gigabitethernet 0/0/1, Description: Verify that the redundant link on the GigabitEthernet 0/0/1 interface is operational and can take over in the event of a failure.